WORKSHEET

Linear transformations

16	6	11		4		2	15	4		12	7	7		9	6	15	13						
8	14	14	10		5	11			6	11	8			3	12	10	1	8	4				

Old proverb

Questions

Express the following linear transformations as matrices:

1 $(x, y) \to (2x + 3y, 3x - 2y)$

2 $(x, y) \to (3x - 2y, 2x + 3y)$

3 $(x, y) \to (5x - 2y, 5x + 2y)$

4 $(x, y) \to (2y - 3x, 2y + 3x)$

5 $(x, y) \to (5y + 2x, 5y - 2x)$

6 $(x, y) \to (5x + 2y, 2y - 5x)$

Find the image of each of the following points under the given linear transformation.

7 $(7, 3)$ under $(x, y) \to (2x + 4y, 3y - 2x)$

8 $(2, -1)$ under $(x, y) \to (3x + y, x + 2y)$

9 $(-3, 5)$ under $(x, y) \to (2x - y, 3x - 2y)$

10 $(2, 6)$ under $(x, y) \to (x + y, y - x)$

Find the image of each of the following geometric shapes under the given linear transformation.

11 $A(2, 3)$, $B(4, 4)$, $C(5, 3)$

under $(x, y) \to (x + 2y, y - 2x)$

12 $A(-1, -2)$, $B(4, 3)$, $C(2, 5)$

under $(x, y) \to (x - 2y, x - y)$

13 $A(2, -4)$, $B(0, -1)$, $C(-1, 2)$

under $(x, y) \to (3x + y, 3x - y)$

14 $A(-1, 4)$, $B(-1, -2)$, $C(3, 5)$, $D(3, 11)$

under $(x, y) \to (2y, y - 2x)$

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15 $A(0, -2), B(4, 1), C(5, 4), D(1, 1)$

under $(x, y) \rightarrow (3x - y, 2x + y)$

16 $A(-4, -3), B(-3, 4), C(4, 3), D(3, -4)$

under $(x, y) \rightarrow (3x, 2x - y)$

Solutions

A $A'(3, 1), B'(-2, 1), C'(-8, -3)$

B $\begin{bmatrix} 5 & -2 \\ 5 & 2 \end{bmatrix}$

D $A'(-12, -5), B'(-9, -10), C'(12, 5), D'(9, 10),$

E $(5, 0)$

G $A'(8, 6), B'(-4, 0), C'(10, -1), D'(22, 5)$

I $\begin{bmatrix} 2 & 5 \\ -2 & 5 \end{bmatrix}$

K $\begin{bmatrix} 2 & 3 \\ 3 & -2 \end{bmatrix}$

L $(26, -5)$

N $A'(8, -1), B'(12, -4), C'(11, -7)$

O $\begin{bmatrix} 2 & 5 \\ -5 & 2 \end{bmatrix}$

P $\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$

R $A'(2, 10), B'(-1, 1), C'(-1, -5)$

S $(8, 4)$

T $\begin{bmatrix} -3 & 2 \\ 3 & 2 \end{bmatrix}$

U $A'(2, -2), B'(11, 9), C'(11, 14), D'(2, 3)$

Y $(-11, -19)$

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